Chapter 3. The Multivariate Normal Distribution

Definition and properties, conditional distributions, best linear prediction, multiple and partial correlation, quadratic forms

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1 Definition

1.1 Non-singular case

The univariate normal density is $f(x) = (2\pi\sigma^2)^{-1/2}\exp[-\frac{1}{2}(x-\mu)\sigma^{-2}(x-\mu)]$. Replacing $(x-\mu)^2/\sigma^2$ by a squared Mahalanobis distance gives the multivariate version.

Definition 1: multivariate normal (non-singular)

For $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\boldsymbol{\Sigma}_{(p\times p)} \succ 0$, $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ means $\mathbf{X}$ has density

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big], \qquad \mathbf{x}\in\mathbb{R}^p,$$

where $|2\pi\boldsymbol{\Sigma}| = (2\pi)^p|\boldsymbol{\Sigma}|$.

$\int f = 1$. Let $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$, so $d\mathbf{y}/d\mathbf{x} = |\boldsymbol{\Sigma}|^{-1/2}$. Then

$$\int f(\mathbf{x})\,d\mathbf{x} = \int\cdots\int (2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}e^{-\mathbf{y}^\top\mathbf{y}/2}|\boldsymbol{\Sigma}|^{1/2}d\mathbf{y} = \prod_{i=1}^{p}\int (2\pi)^{-1/2}e^{-y_i^2/2}dy_i = 1.$$

Equivalently, with $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$ and $\mathbf{z} = \mathbf{Q}^\top(\mathbf{x}-\boldsymbol{\mu})$, the exponent becomes $\sum_i z_i^2/\lambda_i$ and the integral is a product of univariate Gaussian integrals. ■

The substitution shows $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, i.e. $\mathbf{X} \overset{d}{=} \boldsymbol{\Sigma}^{1/2}\mathbf{Y} + \boldsymbol{\mu}$. Hence $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$, $\mathrm{Var}\mathbf{X} = \boldsymbol{\Sigma}$, and

$$U = (\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) = \mathbf{Y}^\top\mathbf{Y} = \textstyle\sum_i Y_i^2 \sim \chi^2(p).$$

1.2 General definition and Cramér–Wold

Definition 2: multivariate normal (singular allowed)

$\mathbf{X} \in \mathbb{R}^p$ has a multivariate normal distribution iff $\mathbf{t}^\top\mathbf{X}$ is univariate normal for every $\mathbf{t} \in \mathbb{R}^p$ (including the point mass $N(c, 0)$).

This definition does not require $\mathrm{Cov}(\mathbf{X})$ to be non-singular: $\mathbf{X}$ is MVN iff *every* projection onto a one-dimensional subspace is normal. It is well defined because, by Cramér–Wold (Ch. 2), the distributions of all $\mathbf{t}^\top\mathbf{X}$ determine that of $\mathbf{X}$.

Example 3: normal marginals do not imply joint normality

$X_1 \sim N(0,1)$ and $X_2 = -X_1$ if $-1 \le X_1 \le 1$, $X_2 = X_1$ otherwise. (a) $X_2 \sim N(0,1)$. (b) (X_1, X_2) is not bivariate normal.

Proof. (a) X_1 is symmetric about 0, so $-X_1 \overset{d}{=} X_1$ on $\{-1 \le X_1 \le 1\}$: $P[X_2 \le x_2] = P[-X_1 \le x_2, |X_1| \le 1] + P[X_1 \le x_2, |X_1| > 1] = P[X_1 \le x_2]$. (b) $X_1 - X_2 = 2X_1$ if $|X_1| \le 1$ and 0 otherwise, so $P(X_1 - X_2 = 0) = P(|X_1| > 1) \approx 0.317 \in (0,1)$: $X_1 - X_2$ is neither a continuous normal nor a point mass. A linear combination is not normal, so (X_1, X_2) is not jointly normal. ■

1.3 Characteristic function

Theorem 4: characteristic function of the MVN

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succeq 0$, then $\varphi_{\mathbf{X}}(\mathbf{t}) = \exp\left(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t}\right)$. If $\boldsymbol{\Sigma} \succ 0$, the density above exists.

Proof. $\mathbf{t}^\top \mathbf{X} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$ and $\varphi_{N(m,v)}(s) = \exp(ism - \frac{1}{2}s^2 v)$, so $\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{X}}(1) = \exp(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$. ■

Since the characteristic function depends only on $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the notation $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is justified under the general definition as well.

2 Geometry: Constant-Density Ellipsoids

Location parameter $\boldsymbol{\mu}$, shape parameter $\boldsymbol{\Sigma} \succ 0$. The contours of constant density are

$$E_c = \{\mathbf{x} \in \mathbb{R}^p : f(\mathbf{x}) = c_0\} = \{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2\},$$

where c is the Mahalanobis distance $d_M(\boldsymbol{\mu}, \mathbf{x})$. With $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\lambda_1 \geq \cdots \geq \lambda_p > 0$, E_c is an ellipsoid centred at $\boldsymbol{\mu}$ with principal axes along $\mathbf{u}_i$ and half-lengths proportional to $\sqrt{\lambda_i}$. If $\boldsymbol{\Sigma} = \mathbf{I}_p$ it is a sphere of radius c.

Example 5: bivariate normal

$\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{R}_{\pi/4} \mathrm{diag}(3, 1) \mathbf{R}_{\pi/4}^\top$: elliptical contours with principal axes at $45°$ and $-45°$ and half-axis ratio $\sqrt{3} : 1$.

3 Linear Transformations, Subsets, Independence, Conditionals

Theorem 6: linear combinations

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \cdots + a_p X_p \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$. Conversely, if $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a}$. Normality via the MGF: $M_{\mathbf{a}^\top \mathbf{X}}(t) = \mathrm{E}e^{t\mathbf{a}^\top \mathbf{X}} = M_{\mathbf{X}}(t\mathbf{a}) = \exp(t\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2}t^2 \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$, the MGF of $N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$. The converse is the general definition. ■

Theorem 7: linear transformations

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{A} \in \mathbb{R}^{q \times p}$ and $\mathbf{d} \in \mathbb{R}^p$, then $\mathbf{A}\mathbf{X} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ and $\mathbf{X} + \mathbf{d} \sim N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. Any $\mathbf{b}^\top(\mathbf{A}\mathbf{X}) = (\mathbf{A}^\top \mathbf{b})^\top \mathbf{X}$ is a linear combination of $\mathbf{X}$, normal by Theorem 3, so $\mathbf{A}\mathbf{X}$ is MVN. $\mathbf{a}^\top(\mathbf{X} + \mathbf{d}) = \mathbf{a}^\top \mathbf{X} + \mathbf{a}^\top \mathbf{d}$ only shifts the mean. ■

Theorem 8: subsets

Partition $\mathbf{X} = \begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$. Then $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$; all subsets of $\mathbf{X}$ are normal.

Proof. Theorem 3 with $\mathbf{A} = [\mathbf{I}_{q\times q}\ \mathbf{0}_{q\times(p-q)}]$. Any subset can be relabelled as $\mathbf{X}_1$. ■

Theorem 9: independence and covariance

(a) If $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$, then $\operatorname{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$. (b) If $\begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix} \sim N_{q_1+q_2}$, then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$. (c) If $\mathbf{X}_1 \sim N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ are independent, then $(\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_{q_1+q_2}\Big(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\Big)$.

Proof. (b) The joint characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\Big\{ i(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2) - \tfrac{1}{2} \begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix}^\top \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix} \begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix} \Big\}.$$

If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ it factorises as $\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$: independence. Conversely, independence forces the factorisation, so the cross term $\mathbf{t}_1^\top \boldsymbol{\Sigma}_{12}\mathbf{t}_2$ must vanish for all $\mathbf{t}$, i.e. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$. (c) reads the factorisation backwards. ■

Remark 10: "uncorrelated ⇒ independent" needs joint normality

In Example 1.2 each marginal is normal but the pair is not jointly normal, so (b) does not apply.

Theorem 11: conditional distribution

If $\boldsymbol{\Sigma}_{22} \succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \ \sim\ N_r(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}).$$

The conditional mean is linear in $\mathbf{x}_2$; the conditional covariance $\boldsymbol{\Sigma}_{11\cdot 2} := \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ does not depend on $\mathbf{x}_2$.

Proof. Define $\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, $\mathbf{X}_2^* = \mathbf{X}_2$, i.e. $\mathbf{X}^* = \mathbf{A}\mathbf{X}$ with $\mathbf{A} = \begin{pmatrix}\mathbf{I}_r & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\\ \mathbf{0} & \mathbf{I}_s\end{pmatrix}$. By Theorem 3 $\mathbf{X}^*$ is MVN and

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \begin{pmatrix}\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}.$$

$\operatorname{Cov}(\mathbf{X}_1^*, \mathbf{X}_2^*) = \mathbf{0}$, so $\mathbf{X}_1^* \perp\!\!\!\perp \mathbf{X}_2$ (Theorem 3(b)) and $\mathbf{X}_1^* \sim N_r(\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{11\cdot 2})$. Since $\mathbf{X}_1 = \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ makes the second term constant without changing the law of $\mathbf{X}_1^*$: $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \overset{d}{=} \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$. ■

Remark 12: residuals, projections and partial variance

$\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ is (up to centring) the residual $\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)$ after projecting $\mathbf{X}_1$ onto the span of $\mathbf{X}_2$. The residual is uncorrelated with $\mathbf{X}_2$, and independent of it in the

Gaussian case. $\sigma^2_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$ (more generally $\boldsymbol{\Sigma}_{11\cdot 2}$) is the **partial variance** of $\mathbf{X}_1$ after removing the effect of $\mathbf{X}_2$, i.e. $\mathrm{Var}(\mathbf{X}_1^*)$.

4 Best Linear Prediction, Multiple and Partial Correlation

Let $\mathbf{X} \in \mathbb{R}^q$ be the predictor and $Y \in \mathbb{R}$ the response, $\mathbf{U} = (\mathbf{X}^\top, Y)^\top \in \mathbb{R}^p$ $(p = q+1)$,

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\mu}_X \\ \mu_Y \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \mathrm{Var}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \sigma_Y^2 \end{pmatrix},$$

with sample counterparts $\bar{\mathbf{u}} = (\bar{\mathbf{x}}^\top, \bar{y})^\top$ and $\mathbf{S} = \begin{pmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & s_Y^2 \end{pmatrix}$.

4.1 Best linear predictor

Theorem 13: best linear predictor

$\mathrm{BLP}(Y \mid \mathbf{X}) := \mathbf{A}^*\mathbf{X} + b^*$, $(\mathbf{A}^*, b^*) = \arg\min_{\mathbf{A}\in\mathbb{R}^{1\times q}, b\in\mathbb{R}} \mathrm{E}\{Y - (\mathbf{A}\mathbf{X} + b)\}^2$. With finite second moments and $\boldsymbol{\Sigma}_{XX} \succ 0$,

$$\mathbf{A}^* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}, \qquad b^* = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X, \qquad \mathrm{BLP}(Y \mid \mathbf{X}) = \mu_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X).$$

Proof. Let $W := Y - \mu_Y - \mathbf{A}^*(\mathbf{X} - \boldsymbol{\mu}_X)$; then $\mathrm{E}W = 0$ and $\mathrm{Cov}(W, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{A}^*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$. For any $(\mathbf{A}, b)$,

$$Y - \mathbf{A}\mathbf{X} - b = W + (\mathbf{A}^* - \mathbf{A})(\mathbf{X} - \boldsymbol{\mu}_X) + \{\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b\},$$

three mutually uncorrelated terms (the last is constant), so

$$\mathrm{E}(Y - \mathbf{A}\mathbf{X} - b)^2 = \mathrm{E}W^2 + (\mathbf{A}^* - \mathbf{A})\boldsymbol{\Sigma}_{XX}(\mathbf{A}^* - \mathbf{A})^\top + (\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b)^2 \geq \mathrm{E}W^2,$$

with equality iff $\mathbf{A} = \mathbf{A}^*$ and $b = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X$. ■

Remark 14: properties of the BLP

$\mathrm{E}[\mathrm{BLP}(Y \mid \mathbf{X})] = \mu_Y$, $\mathrm{Var}[\mathrm{BLP}(Y \mid \mathbf{X})] = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$; the residual $W = Y - \mathrm{BLP}(Y \mid \mathbf{X})$ has $\mathrm{E}W = 0$, $\mathrm{Var}W = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Cov}(W, \mathrm{BLP}) = 0$. Under joint normality $\mathrm{BLP}(Y \mid \mathbf{X}) = \mathrm{E}(Y \mid \mathbf{X})$ (Theorem 3). In block notation $\mathrm{E}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\mu}_2$, $\mathrm{Var}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$.

4.2 Multiple correlation

Definition 15: multiple correlation of Y with $\mathbf{X}$

$$\rho_{\mathrm{mult}} = \mathrm{mcorr}(\mathbf{X}, Y) := \max_{\mathbf{a}\in\mathbb{R}^q\setminus\{\mathbf{0}\}} \mathrm{Corr}(Y, \mathbf{a}^\top\mathbf{X}), \qquad \rho^2_{\mathrm{mult}} = \frac{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}{\sigma_Y^2}.$$

With $\hat{Y} = \mathrm{BLP}(Y \mid \mathbf{X})$, $\rho_{\mathrm{mult}} = \mathrm{Corr}(Y, \hat{Y})$.

Proof. $\mathrm{Corr}(Y, \mathbf{a}^\top \mathbf{X} + b) = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}_{XY}}{\sigma_Y \sqrt{\mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a}}}$. With $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{1/2} \mathbf{a}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top \boldsymbol{\Sigma}_{XY} = \mathbf{b}^\top \boldsymbol{\Sigma}_{XX}^{-1/2} \boldsymbol{\Sigma}_{XY} \le \sqrt{\mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a}} \sqrt{\boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\Sigma}_{XY}},$$

so Corr $\le \sqrt{\boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\Sigma}_{XY}} / \sigma_Y$, with equality iff $\mathbf{a} \propto \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\Sigma}_{XY}$ (the BLP direction). Then $\mathrm{Cov}(Y, \hat{Y}) = \mathrm{Var}(\hat{Y}) = \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\Sigma}_{XY}$ and $\mathrm{Corr}(Y, \hat{Y}) = \sqrt{\mathrm{Var} \hat{Y}} / \sigma_Y$. ■

For two vectors, $\rho(\mathbf{X}_1, \mathbf{X}_2) := \max \mathrm{Corr}(\mathbf{a}^\top \mathbf{X}_1, \mathbf{b}^\top \mathbf{X}_2)$ is the largest canonical correlation (Ch. 8).

4.3 Partial covariance and partial correlation

Partition $\mathbf{X} = (X, \mathbf{Z}^\top)^\top$ with X the predictor of interest and $\mathbf{Z} \in \mathbb{R}^{q-1}$ the others ($\mathrm{Var} \mathbf{Z} \succ 0$), and let $\tilde{X} = X - \mathrm{BLP}(X \mid \mathbf{Z})$, $\tilde{Y} = Y - \mathrm{BLP}(Y \mid \mathbf{Z})$.

Definition 16: partial covariance and partial correlation

$$\sigma_{YX \cdot Z} := \mathrm{Cov}(\tilde{Y}, \tilde{X}) = \mathrm{Cov}(Y, X) - \mathrm{Cov}(Y, \mathbf{Z}) \mathrm{Var}(\mathbf{Z})^{-1} \mathrm{Cov}(\mathbf{Z}, X),$$

$$\rho_{YX \cdot Z} := \frac{\sigma_{YX \cdot Z}}{\sqrt{\mathrm{Var}(\tilde{Y}) \mathrm{Var}(\tilde{X})}} = \mathrm{Corr}(\tilde{Y}, \tilde{X}).$$

In block notation: $\mathrm{pcorr}(\mathbf{X}_1, \mathbf{X}_2; \mathbf{X}_3) = \mathrm{Corr}(\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_3), \mathbf{X}_2 - \mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_3)) = \mathrm{Corr}(\mathbf{X}_1^*, \mathbf{X}_2^*)$ and the partial covariance matrix $\mathrm{pVar}(\mathbf{X}_1; \mathbf{X}_2) = \mathrm{Var}(\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}$.

Partial covariance formula. $\tilde{Y} = Y - \mu_Y - \boldsymbol{\Sigma}_{YZ} \boldsymbol{\Sigma}_{ZZ}^{-1} (\mathbf{Z} - \boldsymbol{\mu}_Z)$ and similarly for $\tilde{X}$. Expanding, $\mathrm{Cov}(\tilde{Y}, \tilde{X}) = \sigma_{YX} - 2 \boldsymbol{\Sigma}_{YZ} \boldsymbol{\Sigma}_{ZZ}^{-1} \boldsymbol{\Sigma}_{ZX} + \boldsymbol{\Sigma}_{YZ} \boldsymbol{\Sigma}_{ZZ}^{-1} \boldsymbol{\Sigma}_{ZZ} \boldsymbol{\Sigma}_{ZZ}^{-1} \boldsymbol{\Sigma}_{ZX} = \sigma_{YX} - \boldsymbol{\Sigma}_{YZ} \boldsymbol{\Sigma}_{ZZ}^{-1} \boldsymbol{\Sigma}_{ZX}$. ■

So the partial covariance matrix of (Y, X) is a Schur complement, and under joint normality it equals the conditional covariance given $\mathbf{Z}$.

Theorem 17: Frisch–Waugh–Lovell theorem

Let $\hat{\beta}_X$ be the coefficient of X in the OLS regression of Y on $(X, \mathbf{Z})$, and let $\tilde{y}_i, \tilde{x}_i$ be the residuals from the OLS regressions of Y and X on $\mathbf{Z}$. Then

$$\hat{\beta}_X = \frac{\sum_i \tilde{x}_i \tilde{y}_i}{\sum_i \tilde{x}_i^2},$$

the slope of the residual-on-residual regression (all regressions with intercept; full-column-rank design).

Proof. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{P}_{[\mathbf{1}, \mathbf{Z}]}$. The normal equations give $\mathbf{y} = \hat{\beta}_X \mathbf{x} + [\mathbf{1}, \mathbf{Z}] \hat{\boldsymbol{\gamma}} + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{x}, \mathbf{1}, \mathbf{Z}$. Multiplying by $\mathbf{M}_Z$ ($\mathbf{M}_Z [\mathbf{1}, \mathbf{Z}] = \mathbf{0}$, $\mathbf{M}_Z \mathbf{e} = \mathbf{e}$) gives $\tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}} + \mathbf{e}$. Multiply by $\tilde{\mathbf{x}}^\top$: $\tilde{\mathbf{x}}^\top \mathbf{e} = \mathbf{x}^\top \mathbf{M}_Z \mathbf{e} = \mathbf{x}^\top \mathbf{e} = 0$, so $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}$. ■

Example 18: mtcars — marginal versus partial correlation

Y = mpg, X = displacement, $\mathbf{Z}$ = (horsepower, weight). The marginal correlation is $\hat{\rho}_{YX} \approx -0.805$, but after adjusting for horsepower and weight $\hat{\rho}_{YX \cdot Z} \approx -0.044$: little linear association remains. The displacement–mpg relation runs mostly through horsepower and weight.

5 Quadratic Forms in Normal Random Vectors

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{A}$, a quadratic form is $Y = \mathbf{X}^\top \mathbf{A}\mathbf{X} = \sum_{j,k} X_j a_{jk} X_k$. We want its distribution and conditions for independence. Special case: $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, $\mathbf{A} = \mathbf{I}_p$ gives $Y = \sum X_i^2 \sim \chi^2(p)$.

Remark 19: matrix algebra recall

$\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$. If $\mathbf{A}$ is symmetric, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$; if it is also idempotent, its eigenvalues are 0 or 1 and $\operatorname{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \operatorname{tr}(\mathbf{A})$.

Theorem 20: chi-square quadratic forms, $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$

Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2\mathbf{I})$ and $\mathbf{A}$ symmetric. Then $Y = \dfrac{\mathbf{X}^\top\mathbf{A}\mathbf{X}}{\sigma^2} \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$.

Proof. ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \operatorname{diag}(1, \dots, 1, 0, \dots, 0)$ (m ones). Let $\mathbf{z} = \frac{1}{\sigma}\boldsymbol{\Gamma}^\top\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}) = N(\mathbf{0}, \mathbf{I}_p)$; then $Y = \mathbf{z}^\top\boldsymbol{\Lambda}\mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$. ($\Rightarrow$) In general $Y = \sum_i \lambda_i z_i^2$ has MGF $\prod_i (1 - 2t\lambda_i)^{-1/2}$; matching $(1-2t)^{-m/2}$ near $t = 0$ forces the nonzero λ_i to be m ones. ■

Simple example: $\mathbf{A} = \operatorname{diag}(1, 0, \dots, 0)$ gives $\mathbf{X}^\top\mathbf{A}\mathbf{X}/\sigma^2 = X_1^2/\sigma^2 \sim \chi_1^2$.

Theorem 21: chi-square quadratic forms, general $\boldsymbol{\Sigma}$

Let $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top\mathbf{A}\mathbf{X} \sim \chi^2(m)$ iff either (1) $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent of rank m, or (2) $\boldsymbol{\Sigma}\mathbf{A}$ is idempotent of rank m.

Proof. By the representation theorem $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{z}$, $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so $Y = \mathbf{z}^\top\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}\mathbf{z}$. By Theorem 5, $Y \sim \chi^2_m$ iff $\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}$ is (symmetric) idempotent of rank m. Now

$$(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2})^2 = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} \iff \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} \iff \mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A},$$

(multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides), and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\boldsymbol{\Sigma})^2 = \mathbf{A}\boldsymbol{\Sigma}$ (multiply on the right by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$). Ranks: $\operatorname{rank}(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\boldsymbol{\Sigma})$. Since $\boldsymbol{\Sigma}\mathbf{A} = (\mathbf{A}\boldsymbol{\Sigma})^\top$, the same holds for $\boldsymbol{\Sigma}\mathbf{A}$. ■

Example 22: distribution of the Mahalanobis distance

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then $(\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2(p)$ ($\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ makes $\mathbf{A}\boldsymbol{\Sigma} = \mathbf{I}_p$ idempotent of rank p). Hence $\{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_p(\alpha)\}$ is an ellipsoid of probability $1-\alpha$.

Theorem 23: independence of linear and quadratic forms

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{A}$ a $p \times p$ symmetric matrix and $\mathbf{B}$ a $k \times p$ matrix. If $\mathbf{B}\mathbf{A} = \mathbf{0}$, then $\mathbf{B}\mathbf{X}$ and $\mathbf{X}^\top\mathbf{A}\mathbf{X}$ are independent.

Proof. If $\mathbf{A}$ is full rank, $\mathbf{B}\mathbf{A} = \mathbf{0}$ forces $\mathbf{B} = \mathbf{0}$: trivial. If $\operatorname{rank}(\mathbf{A}) = m < p$, write $\mathbf{A} = \boldsymbol{\Gamma}\begin{pmatrix}\boldsymbol{\Lambda}_m & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda}_m = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$, all $\lambda_i \neq 0$. From $\mathbf{B}\mathbf{A} = \mathbf{0}$, $\mathbf{B}\boldsymbol{\Gamma}\begin{pmatrix}\boldsymbol{\Lambda}_m & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix} = \mathbf{0}$. Writing $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{C}_1, \mathbf{C}_2]$ gives $\mathbf{C}_1\boldsymbol{\Lambda}_m = \mathbf{0}$, so $\mathbf{C}_1 = \mathbf{0}$ and $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{0}, \mathbf{C}_2]$. Let $\mathbf{z} = \begin{pmatrix}\mathbf{z}_1\\ \mathbf{z}_2\end{pmatrix} = \boldsymbol{\Gamma}^\top\mathbf{X} \sim$

$N_p(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1 \perp\!\!\!\perp \mathbf{z}_2$. Then

$$\mathbf{BX} = \mathbf{B\Gamma z} = \mathbf{C}_2\mathbf{z}_2, \qquad \mathbf{X}^\top \mathbf{AX} = \mathbf{z}^\top \begin{pmatrix} \mathbf{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{z} = \mathbf{z}_1^\top \mathbf{\Lambda}_m \mathbf{z}_1,$$

functions of $\mathbf{z}_2$ and $\mathbf{z}_1$ respectively, hence independent. ■

Remark 24: same argument, two quadratic forms

If $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{AB} = \mathbf{0}$, then $\mathbf{X}^\top \mathbf{AX} \perp\!\!\!\perp \mathbf{X}^\top \mathbf{BX}$. Independence of $\bar{X}$ and S^2 is the case $\mathbf{B} = \frac{1}{n}\mathbf{1}^\top$, $\mathbf{A} = \mathbf{H}_n$ with $\mathbf{BA} = \mathbf{0}$; the multivariate version is in Ch. 4.